

Main Content

1. Inverse Laplace Transform
2. Single Real Poles
3. Complex Conjugate Poles
4. Multiple-Order Poles

Key Focus: Partial fraction expansion method for inverse Laplace transforms

Critical Challenges: Handling complex conjugate poles and repeated roots

1. Inverse Laplace Transform

1. **Complex Contour Integration via Definition** - Uses the residue theorem to evaluate contour integrals
2. **Partial Fraction Expansion — Table Lookup Method (Key Technique)**
 - **Decomposes $F(s)$ into simpler fractions, then matches them to time-domain functions via standard tables**
3. **Computer-Aided Solutions**
 - **Implemented in tools like MATLAB**

5. Inverse Laplace Transform

(1) Partial Fraction Expansion

$$\mathcal{L}[\delta(t)] = 1$$

$$\mathcal{L}[\delta^{(n)}(t)] = s^n$$

Standard Signal
Transform Table

$$\mathcal{L}[u(t)] = \frac{1}{s} \quad \mathcal{L}[t] = \frac{1}{s^2} \quad \mathcal{L}[t^n] = \frac{n!}{s^{n+1}}$$

$$\mathcal{L}[\cos(\omega_0 t)u(t)] = \frac{s}{s^2 + \omega_0^2} \quad \mathcal{L}[\sin(\omega_0 t)u(t)] = \frac{\omega_0}{s^2 + \omega_0^2}$$

$$\mathcal{L}[e^{-\alpha t}u(t)] = \frac{1}{s + \alpha} \quad \mathcal{L}[e^{-\alpha t}\cos(\omega_0 t)u(t)] = \frac{s + \alpha}{(s + \alpha)^2 + \omega_0^2}$$

$$\mathcal{L}[e^{-\alpha t}\sin(\omega_0 t)u(t)] = \frac{\omega_0}{(s + \alpha)^2 + \omega_0^2}$$

5. Inverse Laplace Transform

(1) Partial Fraction Expansion

Typically, $F(s)$ is expressed as a rational function:

$$F(s) = \frac{B(s)}{A(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_1 s + b_0}{a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0}$$

a_i, b_i are real coefficients, m, n are positive integers

If $m < n$, $F(s)$ is a proper rational function

Factorization

$$F(s) = \frac{B(s)}{A(s)} = \frac{b_m (s - z_1)(s - z_2) \cdots (s - z_m)}{a_n (s - p_1)(s - p_2) \cdots (s - p_n)}$$

Zeros: $z_1, z_2, z_3 \cdots z_m$ are roots of $B(s) = 0$

Poles: $p_1, p_2, p_3 \cdots p_n$ are roots of $A(s) = 0$

5. Inverse Laplace Transform

(1) Partial Fraction Expansion

Step-by-Step Procedure:

- Identify Poles of $F(s)$
- Expand $F(s)$ into Partial Fractions
- Inverse Transform $f(t)$ via Laplace Table

5. Inverse Laplace Transform

(1) Partial Fraction Expansion

Proper Rational Function: $F(s) = \frac{B(s)}{(s - p_1)(s - p_2) \cdots (s - p_n)}$

Case 1: Distinct Real Poles

$p_1, p_2, p_3 \dots p_n$ are distinct real roots

$$\frac{1}{(s - p_1)(s - p_2)} = \frac{k_1}{s - p_1} + \frac{k_2}{s - p_2}$$

$$\frac{k}{s - p} \leftrightarrow e^{pt} u(t)$$

(1) Partial Fraction Expansion

Case 2: Complex Conjugate Poles

Distinct complex conjugate poles (real and imaginary parts affect time-domain behavior)
Treat conjugate pairs together to avoid complex arithmetic

$$\frac{1}{(s-p)(s-p^*)} = \frac{1}{s^2 - (\textcolor{red}{p} + \textcolor{red}{p}^*)s + \textcolor{red}{p}p^*}$$

Case 3: Repeated Poles

$p_1, p_2, p_3 \dots p_n$ multiple identical roots

$$\frac{1}{(s-p_1)^2 (s-p_3)} = \frac{\textcolor{red}{k}_1}{s-p_1} + \frac{k_2}{(s-p_1)^2} + \frac{k_3}{s-p_3}$$

5. Inverse Laplace Transform

Example 1 First-Order Real Poles

$$F(s) = \frac{s+1}{s^2+5s+6}$$

Step 1: Find poles

$$F(s) = \frac{s+1}{(s+2)(s+3)}$$

Step 2: Partial fraction expansion

$$F(s) = \frac{k_1}{s+2} + \frac{k_2}{s+3}$$

Step 3: Determine coefficients

$$F(s) = \frac{-1}{s+2} + \frac{2}{s+3}$$

Step 4: Inverse transform

Using the Laplace pair $\mathcal{L}\left[\mathbf{e^{-\alpha t} u(t)}\right] = \frac{\mathbf{1}}{\mathbf{s + \alpha}}$

$$f(t) = \left(-e^{-2t} + 2e^{-3t}\right)\mathbf{u(t)}$$

5. Inverse Laplace Transform

How to Determine Coefficients k_1, k_2 ?

$$F(s) = \frac{s+1}{(s+2)(s+3)} = \frac{k_1}{s+2} + \frac{k_2}{s+3}$$

Heaviside's "Cover-up" Method

Step 1: Find k_1 , multiply both sides by $s+2$ (to isolate k_1)

$$\text{Right Side} = k_1 + \frac{k_2}{s+3}(s+2)$$

Let $s = -2$, right side = k_1

$$\begin{aligned} \text{Left Side} &= (s+2)F(s)\big|_{s=-2} \\ &= (s+2) \frac{s+1}{(s+2)(s+3)} \bigg|_{s=-2} = -1 \end{aligned}$$

Similarly $k_2 = (s+3)F(s)\big|_{s=-3} = 2$

5. Inverse Laplace Transform

Example 2 First-Order Complex Conjugate Poles

Find the inverse Laplace transform of $F(s) = \frac{s^2 + 3}{(s + 2)(s^2 + 2s + 5)}$

(1) Identify Poles (preserve conjugate terms):

$$F(s) = \frac{s^2 + 3}{(s + 2) \left[(s + 1)^2 + 4 \right]}$$

$$\begin{cases} \mathcal{L} \left[e^{-\alpha t} \cos(\omega_0 t) u(t) \right] = \frac{s + \alpha}{(s + \alpha)^2 + \omega_0^2} \\ \mathcal{L} \left[e^{-\alpha t} \sin(\omega_0 t) u(t) \right] = \frac{\omega_0}{(s + \alpha)^2 + \omega_0^2} \end{cases}$$

(2) Partial Fraction Expansion:

Key Laplace Pairs

$$F(s) = k_1 \frac{1}{s + 2} + k_2 \frac{s + 1}{(s + 1)^2 + 2^2} + k_3 \frac{2}{(s + 1)^2 + 2^2}$$

5. Inverse Laplace Transform

Example 2 First-Order Complex Conjugate Poles

$$\begin{aligned} F(s) &= \frac{s^2 + 3}{(s + 2)(s^2 + 2s + 5)} \\ &= \frac{k_1}{s + 2} + \frac{k_2(s + 1)}{(s + 1)^2 + 4} + \frac{k_3 2}{(s + 1)^2 + 4} \end{aligned}$$

(3) Coefficient Calculation

$$k_1 = (s + 2)F(s)\big|_{s=-2} = \frac{7}{5}$$

$$F(s) = \frac{7}{5} \frac{1}{s + 2} + \frac{k_2(s + 1)}{(s + 1)^2 + 4} + \frac{k_3 2}{(s + 1)^2 + 4}$$

Combine right-hand side terms and equate numerators

5. Inverse Laplace Transform

Example 2 First-Order Complex Conjugate Poles

$$F(s) = \frac{\frac{7}{5}(s^2 + 2s + 5) + k_2(s+1)(s+2) + k_3 2(s+2)}{(s+2)(s^2 + 2s + 5)}$$

$$\begin{cases} s^2, \text{ coefficient: } 1 = \frac{7}{5} + k_2 \\ s, \text{ coefficient: } 0 = \frac{14}{5} + 3k_2 + 2k_3 \end{cases} \quad \text{thus} \quad \begin{cases} k_2 = -\frac{2}{5} \\ k_3 = -\frac{4}{5} \end{cases}$$

(4) Inverse transform

$$f(t) = \frac{7}{5} e^{-2t} u(t) + e^{-t} \left[-\frac{2}{5} \cos(2t) - \frac{4}{5} \sin(2t) \right] u(t)$$

5. Inverse Laplace Transform

Example 3 Multiple-Order Poles

$$F(s) = \frac{s^2}{(s+2)(s+1)^2} = \frac{k_1}{s+2} + \frac{k_2}{s+1} + \frac{k_3}{(s+1)^2}$$

k_1 Coefficient for simple pole, k_3 Highest-order coefficient for repeated pole

$$k_1 = (s+2) \frac{s^2}{(s+2)(s+1)^2} \Big|_{s=-2} = 4$$

$$k_3 = (s+1)^2 \frac{s^2}{(s+2)(s+1)^2} \Big|_{s=-1} = 1$$

How to determine k_2 ?

5. Inverse Laplace Transform

Example 3 Multiple-Order Poles, Solving for k_2

$$F(s) = \frac{s^2}{(s+2)(s+1)^2} = \frac{k_1}{s+2} + \frac{k_2}{s+1} + \frac{k_3}{(s+1)^2}$$

Isolate k_2 by eliminating other terms

Multiply by $(s+1)^2$

$$\text{Right} = (s+1)^2 \frac{k_1}{s+2} + k_2(s+1) + k_3$$

Differentiate Both Sides with respect to s

$$\text{Right} = \left[(s+1)^2 \left(\frac{k_1}{s+2} \right)' + 2(s+1) \frac{k_1}{s+2} \right] + k_2 + 0$$

Evaluate at $s = -1$, $\text{Right} = k_2$

$$k_2 = \text{left} = \frac{d}{ds} \left[\frac{s^2}{s+2} \right]_{s=-1} = \frac{s^2 + 4s}{(s+2)^2} \Big|_{s=-1} = -3$$

5. Inverse Laplace Transform

Example 3 Multiple-Order Poles

$$F(s) = \frac{4}{s+2} + \frac{-3}{s+1} + \frac{1}{(s+1)^2}$$

$$\frac{1}{s^{n+1}} \rightarrow \frac{1}{n!} t^n u(t)$$

$$\frac{1}{(s+\alpha)^{n+1}} \rightarrow e^{-\alpha t} \frac{1}{n!} t^n u(t)$$

$$f(t) = \mathcal{L}^{-1}[F(s)] = \left(4e^{-2t} - 3e^{-t} + te^{-t} \right) u(t)$$

Summary

Partial Fraction Expansion

$$F(s) = \frac{s+1}{s^2+5s+6}$$

$$F(s) = \frac{s+\gamma}{(s+\alpha)^2+\beta^2}$$

$$F(s) = \frac{s^2}{(s+2)(s+1)^2}$$

- Identify Poles of $F(s)$
- Expand $F(s)$ into Partial Fractions
- Inverse Transform $f(t)$ via Laplace Table

Main Content

- Introduction to Analyzing Dynamic Circuits Using Laplace Transform Method
- s-Domain Equivalent Models of Circuits
- Dynamic Circuit Analysis

1. Introduction to Analyzing Dynamic Circuits

- ✓ First, establish the s-domain equation (two approaches)
 - Formulate the time-domain differential equation, then apply the differentiation and integration properties to obtain the Laplace transform (**this method is not used here**)
 - **Directly construct the algebraic equation based on the s-domain model of the circuit**
- ✓ Solve the s-domain equation

$$F(s) \rightarrow f(t)$$

to obtain the time-domain solution

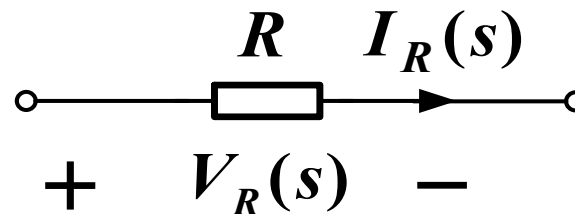
2. s-Domain Equivalent Models of Circuits

Circuit **components** are represented by their **complex frequency-domain** models, and **excitations** are replaced by their **Laplace** transforms, thereby forming the **s-domain model** of the circuit

(1) s-Domain Model of a Resistor

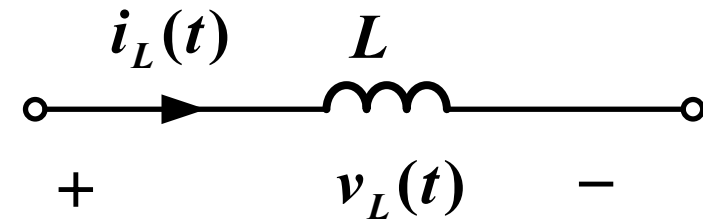
Time-domain relationship: $v_R(t) = Ri_R(t)$

Transform-domain relationship: $V_R(s) = RI_R(s)$



(2) s-Domain Model of an Inductor

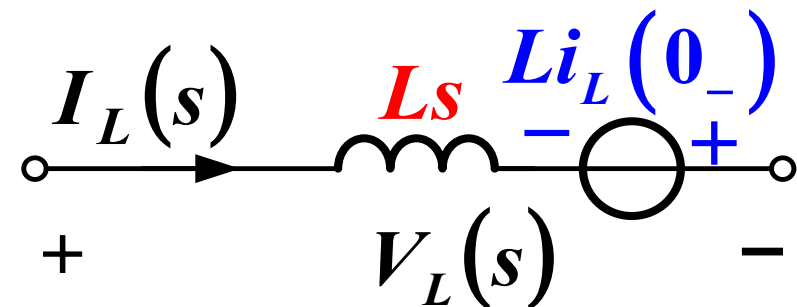
Time-domain
relationship: $v_L(t) = L \frac{di_L(t)}{dt}$



Let $\begin{cases} \mathcal{L}[i_L(t)] = I_L(s) \\ \mathcal{L}[v_L(t)] = V_L(s) \end{cases}$

Applying the time-domain differentiation property of the Laplace transform:

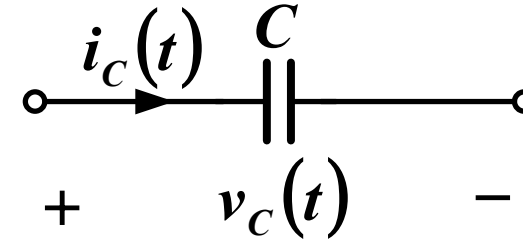
$$\begin{aligned} V_L(s) &= L[sI_L(s) - i_L(0_-)] \\ &= LsI_L(s) - Li_L(0_-) \end{aligned}$$



(3) s-Domain Model of a Capacitor

Time-domain relationship: $v_C(t) = \frac{1}{C} \int_{-\infty}^t i_C(\tau) d\tau$

Let $\begin{cases} \mathcal{L}[i_C(t)] = I_C(s) \\ \mathcal{L}[v_C(t)] = V_C(s) \end{cases}$



Using the integration property of the unilateral Laplace transform:

$$V_C(s) = \frac{1}{C} \left[\frac{\textcolor{red}{I}_C(s)}{\textcolor{red}{s}} + \frac{i_C^{(-1)}(0_-)}{s} \right]$$

s-Domain relationship:

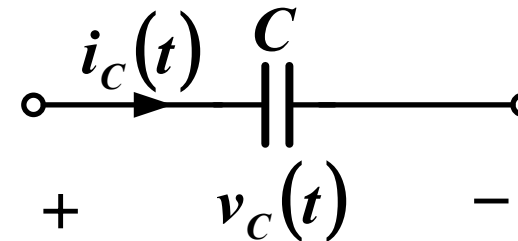
$$V_C(s) = \frac{1}{sC} \textcolor{red}{I}_C(s) + \frac{\textcolor{blue}{v}_C(0_-)}{s}$$

$$\begin{aligned} \frac{1}{C} i_C^{(-1)}(0_-) &= \frac{1}{C} \int_{-\infty}^{0_-} i_C(\tau) d\tau \\ &= v_C(0_-) \end{aligned}$$

(3) s-Domain Model of a Capacitor

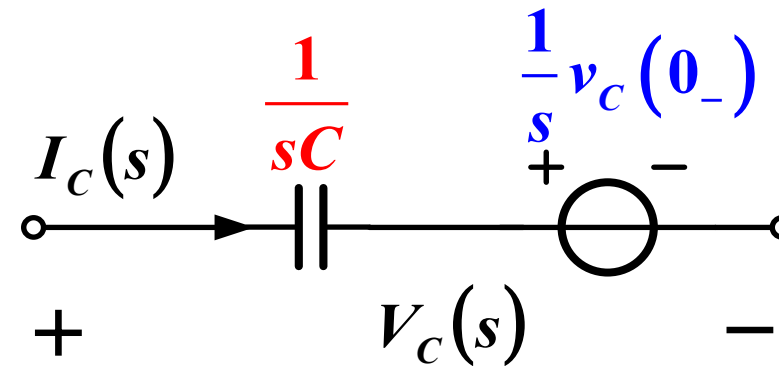
Time-domain relationship:

$$v_C(t) = \frac{1}{C} \int_{-\infty}^t i_C(\tau) d\tau$$



s-Domain relationship:

$$V_C(s) = \frac{1}{sC} I_C(s) + \frac{v_C(0_-)}{s}$$



3. Dynamic Circuit Analysis

(1) Complex Frequency-Domain Forms of Circuit Theorems

$$i(t) \xleftrightarrow{LT} I(s), \quad \text{KCL} : \sum i(t) = 0 \rightarrow \sum I(s) = 0$$

$$v(t) \xleftrightarrow{LT} V(s) \quad \text{KVL} : \sum v(t) = 0 \rightarrow \sum V(s) = 0$$

All standard linear circuit analysis techniques remain valid in the s-domain

3. Dynamic Circuit Analysis

(2) Steps for Solving Responses

- Draw the $t=0$ equivalent circuit to determine the initial values of state variables $v_C(0_-)$ and $i_L(0_-)$
- Construct the s-domain equivalent model
- Formulate s-domain equations (algebraic equations)
- Solve the s-domain equations to obtain the Laplace transform of the response $V(s)$ or $I(s)$
- Perform the inverse Laplace transform to find $v(t)$ or $i(t)$

6. LT Analysis of Dynamic Circuits

Example 2

The circuit shown below has zero initial state. At $t = 0$, switch S closes, connecting to DC power source E . Find the current $i(t)$

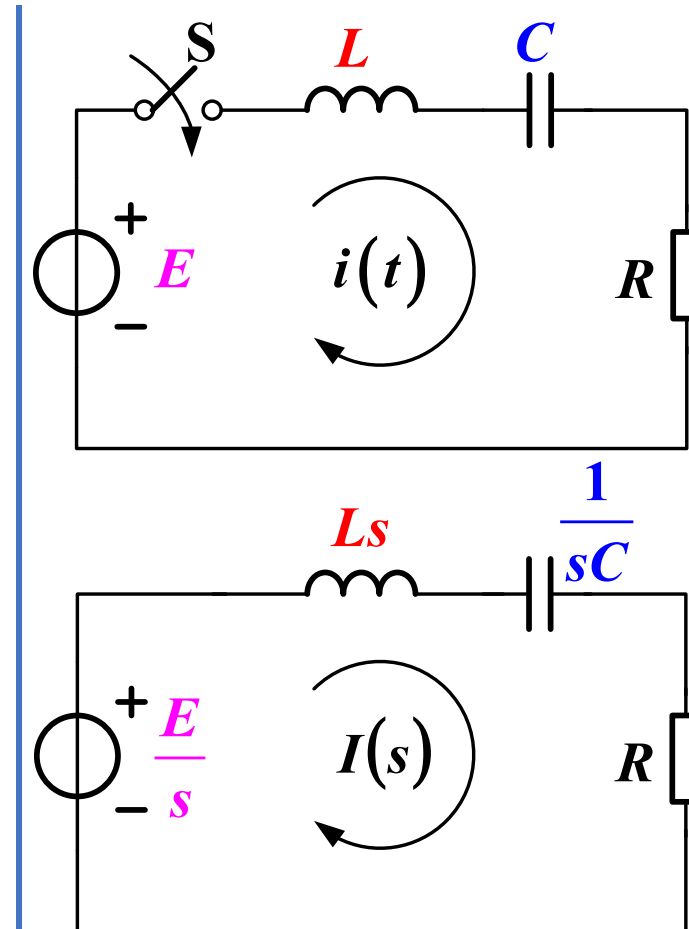
(1) Zero initial conditions:

$$\Rightarrow i_L(0_-) = 0\text{A}, v_C(0_-) = 0\text{V}$$

(2) Draw the s-domain equivalent model for $t > 0$

(3) Write KVL equation:

$$\frac{E}{s} = LsI(s) + \frac{1}{Cs}I(s) + RI(s)$$



6. LT Analysis of Dynamic Circuits

Example 2

$$\frac{E}{s} = LsI(s) + \frac{1}{Cs}I(s) + RI(s)$$

Obtain

$$I(s) = \frac{E}{s\left(Ls + R + \frac{1}{sC}\right)} = \frac{E}{L} \frac{1}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

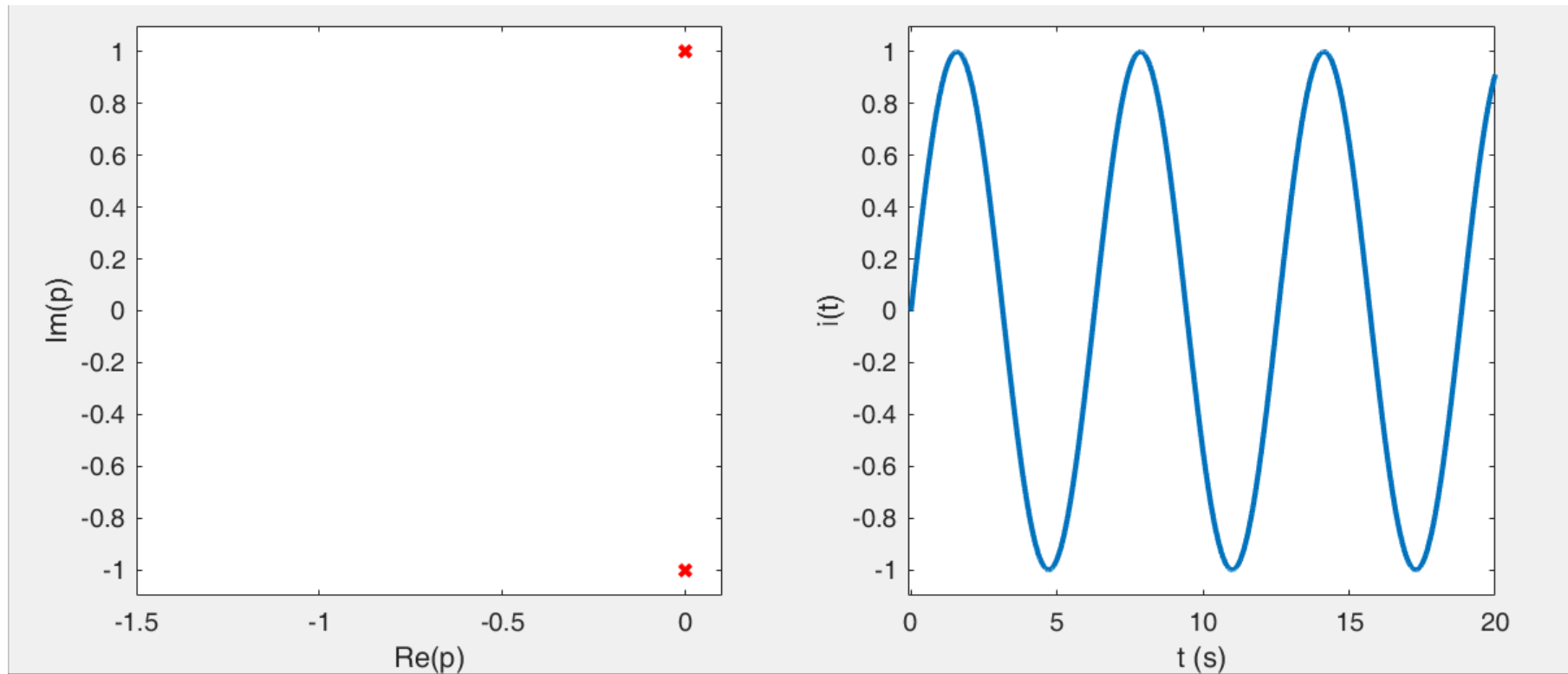
Poles can be expressed as

$$p_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

$$p_2 = -\frac{R}{2L} - \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

6. LT Analysis of Dynamic Circuits

Example 2

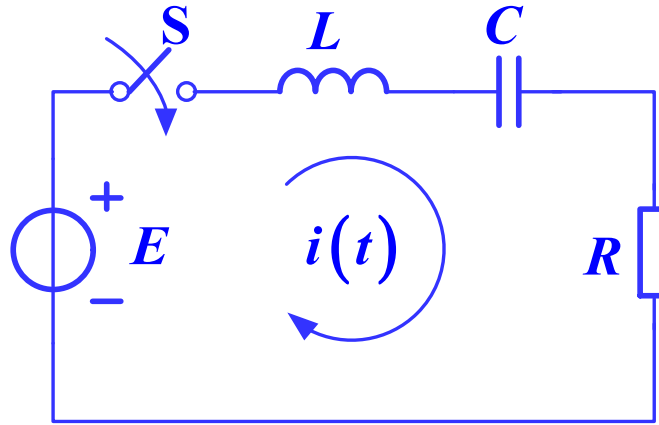


$$p_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

$$p_2 = -\frac{R}{2L} - \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

6. LT Analysis of Dynamic Circuits

Example 2

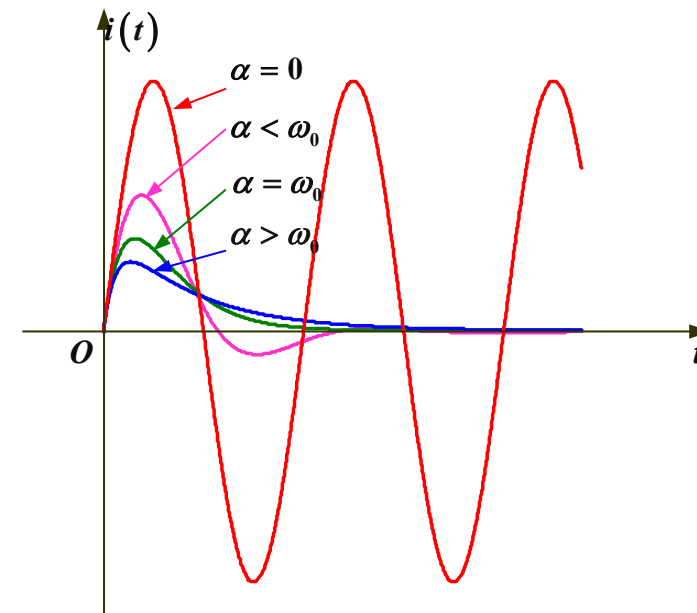


$$i(t) = \frac{E}{L(p_1 - p_2)} (e^{p_1 t} - e^{p_2 t}) u(t)$$

$$\text{let } \alpha = \frac{R}{2L}, \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\text{Then } p_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}, \quad p_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

Real part Imaginary part



1. **Undamped** oscillation: constant amplitude oscillation
2. **Underdamped** oscillation: Exponentially decaying oscillation
3. **Critical damping**: Fastest response without overshoot
4. **Overdamping**: Slow non-oscillatory response

Summary

Analyzing Dynamic Circuits Using Laplace Transform Method

- ✓ **Directly construct the algebraic equation based on the s-domain model of the circuit**
- ✓ **Solve the s-domain equation**
- ✓ **$F(s) \rightarrow f(t)$, to obtain the time-domain solution**